



Omii Chauhan

Roll No.: 23MMB0A31

Bachelor of Technology — Metallurgy and Materials Science

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Education

National Institute of Technology, Warangal	2023 – 2027
Bachelor of Technology — Metallurgy and Materials Science	
Telangana State Board of Intermediate Education	2022
Class XII (Higher Secondary) — 96%	
Central Board of Secondary Education	2020
Class X (Secondary School) — 93%	

Experience

ADEN HQ (hive) — Multi-Agent Harness for Production AI (10k+ Stars)	Feb 2026 – Present
Core Framework Contributor Remote	
- Concurrency & Resilience: Engineered a per-key asynchronous locking strategy to resolve a critical TOCTOU race condition, and refactored the file-spillover and session checkpointing pipelines to isolate corrupted states, preventing silent execution failures on resume. - Performance Engineering: Refactored legacy synchronous tool integrations (Exa, Apollo) to native asynchronous execution using <code>httpx.AsyncClient</code>, eliminating event-loop starvation. - Advanced Swarm Consensus: Implemented the core engine for the Neutrosophic Scoring Module (RFC #7179), introducing multi-dimensional confidence logic for swarm decision-making. - Security & Protocol Hardening: Patched critical directory traversal vulnerabilities across 13 routing endpoints and engineered a custom DNSSEC validation pipeline bypassing system resolvers to enforce Authenticated Data (AD) verification.	
Rajputana Vehicles Private Limited	May 2025 – June 2025
Data Analyst Intern Madhya Pradesh, India	
- Developed an automated ETL data pipeline using Python (Pandas) to process raw business data, eliminating 10+ hours/week of manual reporting. - Architected customer segmentation models utilizing K-Means clustering on 50k+ transaction records, improving campaign targeting precision by 15%. - Designed interactive KPI dashboards in Power BI to track EV adoption rates and inventory turnover, enabling data-backed strategic decisions for senior management.	

Technical Projects

FinTech AI: Credit Risk Prediction Engine GitHub Live Demo	2024 – 2025
Real-time loan default prediction using ensemble ML models	
Stack: Python, XGBoost, Random Forest, Scikit-learn, Docker, FastAPI - Developed an end-to-end ML pipeline (Random Forest & XGBoost ensemble) to predict borrower default risk, encompassing robust feature engineering and probability calibration. - Engineered a containerized batch processing engine (Docker) capable of scoring 10,000+ loan applications asynchronously via FastAPI workers, computing Exposure at Default (EAD) and Loss Given Default (LGD) metrics in under 5 seconds.	
AI Multi-Strategy Fund Platform GitHub	2025
Autonomous agent-driven investment lifecycle platform	
Stack: Python, LangGraph, FastAPI, Next.js 14, Recursive Risk Parity, Redis - Built a multi-agent orchestration workflow using LangGraph and Redis queues, where independent agents generated allocation signals and portfolio summaries asynchronously. - Implemented hierarchical risk parity-based allocation logic to actively rebalance portfolios using historical market data.	
Visual Workflow Builder GitHub Live Demo	2026
Production-grade node-based pipeline editor with real-time DAG validation	
Stack: React, React Flow, Zustand, FastAPI (Python), REST APIs, CORS - Architected an extensible BaseNode component abstraction, reducing UI boilerplate by 40% and allowing rapid deployment of 5+ custom node types (LLM, Database, File, Timer, Email). - Engineered dynamic variable parsing with custom regex to dynamically render input handles on text nodes, backed by a Zustand global state machine synchronization pipeline. - Developed a FastAPI backend using Kahn's Algorithm (Topological Sort) to perform real-time Directed Acyclic Graph (DAG) validation, cycle detection, and pipeline analysis in $O(V + E)$ complexity.	

Technical Skills

Languages: Python (asyncio), SQL, C++, TypeScript
Data & ML: Pandas, NumPy, Scikit-learn, XGBoost, TensorFlow, Feature Engineering
Backend & Infrastructure: FastAPI, PostgreSQL, Snowflake, Redis, Docker, AWS, DNSSEC
Frontend & Tools: React, Next.js, Git, CI/CD, DevOps

Leadership & Achievements

Leadership: Head of PR & Sponsorship, Technozion & Springspree (Managed 50+ students & corporate sponsors).
Certifications & Simulations: HackerRank SQL (Advanced), J.P. Morgan Quantitative Research, EA Software Engineering.